

HelixCode — Feature Status

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Comprehensive per-feature status ledger (§11.4.153) for the **HelixCode** stack at release `codeserver-1.0.0-dev-0.0.3`, extended with the `post-0.0.3` feature additions landed by commits `8351e4c` (login-form clipboard buttons + UI-driven marketplace journey) and `97a8c69` (VS Code Dark default theme + popular-extensions matrix), plus the now-**LIVE** rootless-Quadlet edge. Every user-facing capability is enumerated with the §11.4.153 field set — Component / Feature / Category / Implementation (the real file) / Wiring (how it is genuinely end-user-reachable, §11.4.108) / Real-use / Tests-coverage (§11.4.169) / Validation / Video-confirmation. Validation vocabulary (§11.4.45): **PASS · SKIP · OPERATOR-BLOCKED · OPERATOR-ATTENDED · PENDING**. Autonomous captured evidence lives under the git-ignored `qa-results/` tree (regenerated by re-running the suite, §11.4.77); the release-gate aggregate is `qa-results/run_all/20260701T151300Z-3182121/` (`summary.txt`), with per-suite evidence under `qa-results/tests/<suite>/`.

Reconciliation basis (§11.4.6). This ledger reconciles against `README.md`, `docs/guides/AUTH.md`, and `docs/changelogs/codeserver-1.0.0-dev-0.0.3.md`. The real-account SSH-key auth surface is **LIVE-validated** this release — release-gate run = **23/23 suites PASS, 0 FAIL** (`summary.txt`), Go gate = **70 tests, -race, 81.8% coverage**. Rows that were "in progress / PENDING" in Revision 1 are now PASS with cited captured evidence.

Components

- **Caddy edge** — the only containerized component (rootless Podman, §11.4.76/§11.4.161). Terminates TLS on `:52443` (HTTP `:52080`→HTTPS 301), speaks HTTP/3 (QUIC), compresses with Brotli (custom xcaddy build), gates every request via `forward_auth` → `helix-auth`, blocks `/proxy`, strips client X-Helix-User. `deploy/Caddyfile`, `deploy/caddy/Dockerfile`.

- **helix-auth** — host-native **Gin** forward-auth gate (systemd --user, loopback 127.0.0.1:8081, **non-root**). Issues + verifies the SSH-key challenge (ssh-keygen -Y verify vs the account's authorized_keys), issues the signed session cookie, **fails closed**, stores no password. services/auth_gate/.
- **code-server** — browser VS Code, **host-native as the real user** (milosvasic), --auth none on loopback 127.0.0.1:8080 (auth enforced upstream at the gate/edge), projects served read-write, Explorer defaults to \$PROJECTS_ROOT. deploy/systemd/helix-code-server.service.
- **Operator scripts** — install / setup / start / stop / status / doctor + install-auth.sh (tarball code-server + units), install-edge-boot.sh (edge Quadlet), harden-loopback.sh (UID-scoped loopback firewall).
- **Boot-survival** — rootless systemd --user units (linger) + the edge Quadlet.

Feature ledger (§11.4.153)

Component	Feature	Category	Implementation
Caddy edge	TLS HTTPS edge (self-signed LAN, per-boot)	edge/tls	deploy/Caddyfile, deploy/c Dockerfile
Caddy edge	HTTP→HTTPS redirect (:52080→:52443, 301)	edge/tls	deploy/Caddyfile
Caddy edge	HTTP/3 (QUIC) + Brotli compression	edge/perf	deploy/Caddyfile (encode zstd br gzip), deploy/caddy/Dockerfile (xcaddy-)
Caddy edge	forward_auth gate wiring (every req → helix-auth /auth)	edge/auth	deploy/Caddyfile forward_a
Caddy edge	/proxy blocked (403) + client X-Helix-User stripped	edge/security	deploy/Caddyfile
Caddy edge	Let's Encrypt ACME issue/renew/rotation (local Pebble CA)	edge/tls	deploy/.env TLS_MODE=letsencrypt, deploy/Caddyfile
Caddy edge	Real public Let's Encrypt (public domain / DNS-01)	edge/tls	docs/guides/TLS.md
Caddy edge	Edge reboot-persistence (rootless Quadlet)	edge/boot	deploy/quadlet/helixcode-caddy.container, scripts/install-edge-boot.sh

Component	Feature	Category	Implementation
helix-auth	SSH-key challenge-response login (verify vs authorized_keys, no password, nothing stored)	auth	services/auth_gate/(challenge.go, verifier.go, cookie.go, server.go), deployed to systemd/helix-auth.service
helix-auth	Login redirect for browser navigations (303→/login; non-HTML/non-GET → 401)	auth	services/auth_gate/server.go, handleAuth (commit 75e2d9)
helix-auth	Auth fails CLOSED (gate down/errored → denied, never bypassed)	auth/security	services/auth_gate/server.go, deploy/Caddyfile
helix-auth	Hardened session cookie (__Host- + HMAC + HttpOnly + Secure + SameSite=Strict, regen on login)	auth/security	services/auth_gate/cookie.go
helix-auth	Rate-limit + CSRF double-submit + verify-spawn ceiling (DoS-hardened)	auth/security	services/auth_gate/ratelimit.go, server.go
helix-auth	Security hardening (server-pinned principal CVE-2026-35414, key-type allow-list, OpenSSH floor, exec/temp hygiene)	auth/security	services/auth_gate/verifier.go, sshversion.go, config.go
helix-auth	Login-form copy/paste clipboard buttons (accessible copy sign-cmd + challenge, paste recognises armored SSH signature, multi-sig picker, XSS-safe, progressive-enhancement)	auth/ux	services/auth_gate/assets/login_enhance.js, services/auth_gate/server.go (commit 8351e4c)

Component	Feature	Category	Implementation
code-server	Host-native editor served (authed 200, x-Helix-User: milosvasic)	editor	deploy/systemd/helix-code-server.service
code-server	Tarball install (~/.local/lib, non-root; npm fallback), pinned 4.117.0	editor/install	scripts/install-auth.sh
code-server	Projects served read-write; Explorer defaults to \$PROJECTS_ROOT (convenience view, NOT a jail)	editor	deploy/systemd/helix-code-server.service, deploy/.env PROJECTS_ROOT
code-server	SSH-key git from the integrated terminal (host ~/.ssh)	editor/env	host-native as real user
code-server	.bashrc / profile sourced in fresh terminal	editor/env	host-native as real user
code-server	Extension (plugin) install + use from Open VSX marketplace	editor/extensions	tests/types/extensions_auth_bank tests/banks/helixcode extensions.yaml
code-server	VS Code Dark enforced as DEFAULT theme ("Visual Studio Dark", every fresh install)	editor/ux	deploy/code-server/settings.default.json (commit 97a8c69)
code-server	Popular extensions install + loadable +		tests/types/extensions_popular_auth.s

Component	Feature	Category	Implementation
	persist across restart + config round-trip (Open VSX)	editor/extensions	bank tests/banks/helixcode-extensions.yaml
code-server	In-browser editing (type/save via UI, file tree, command palette)	editor/ui	code-server workbench
code-server	UI-driven marketplace journey (open Extensions view → search Open VSX → click real Install button, pixel+OCR)	editor/ui	tests/types/extensions_ui_auth.sh, bank tests/banks/helixcode-extensions-ui.yaml (commit 8351e4c)
code-server	file-watcher fix (files.watcherExclude seeded to volume)	editor/env	persistent settings volume
Operator scripts	Loopback firewall (UID-scoped DROP to	security/host	scripts/harden-loopback.s

Component	Feature	Category	Implementation
	:8080 for non-account UIDs)		
Operator scripts	Boot-survival systemd -- user units (linger, WantedBy=default.target)	boot	scripts/install-auth.sh, d systemd/*.service
Stack	Resilience (flood / kill- recover / fd pressure)	resilience	tests/types/ {load_auth,stress_chaos_a
Stack	Performance / memory bounds (p50/p95/p99, RSS < 60%)	performance/ memory	tests/types/ {benchmark_auth,memory_au
Stack	No secret leak / rootless	security	tests/types/security_auth

Release-gate run (§11.4.169)

Aggregate qa-results/run_all/20260701T151300Z-3182121/summary.txt
— **23/23 PASS, 0 FAIL:**

- **Auth surface (LIVE, real id_ed25519):** e2e_auth 5/5 · security_auth 5/5 · challenges_auth 5/5 · concurrency_auth 5/5 · helixqa_auth 4/0/1 · load_auth 3/0/1 · memory_auth 2/2 · benchmark_auth 4/4 · stress_chaos_auth 5/0/1 (live kill+recover, fail-closed verified).
- **Core:** unit 5/5 · race 2/2 (-race) · load 3/3.
- **Go gate: 70 tests, -race, 81.8% coverage.**
- **Legacy (retired container+password model):** integration, e2e, security, tls_letsencrypt, full_automation, concurrency, memory, benchmark, helixqa, challenges each report an honest **SKIP** (topology_unsupported, "superseded by <name>_auth") — no false-FAIL (§11.4.1).
- **Extension surface (separate live run):** extensions_auth 5/5 checks captured (qa-results/tests/extensions_auth/20260701T162817Z-4080597).
- **Post-0.0.3 suites (registered in run_all_types.sh, separate captured runs):** login_ui_auth LIVE 3/3 (L2 node 8/8 + L3 CDP pixel proof ok:true) · theme_default_auth 3/3 config · theme_visual_auth §11.4.170 rendered-pixel proof (dark meanLum 44.3 vs light control 228.5) · extensions_popular_auth install proven live (4/5 + direct 8-ext probe; network-gated) · extensions_ui_auth 5/1 UI-driven (browse→search→Install-click; operator_attended install-completion).

Honest boundaries (§11.4.6)

- **No §11.4.153 real-use VIDEO exists yet** for any HelixCode feature — every row's **Video-confirmation is PENDING**. The autonomous captured evidence backing each PASS is the qa-results/** log + .txt artefact cited in the row (§11.4.69/ §11.4.107), NOT a recorded video. Producing the per-feature real-use videos is a tracked gap; nothing here is claimed video-confirmed until the recording lands.
- **Editor "jail" is cosmetic**. The Explorer defaults to \$PROJECTS_ROOT, but the integrated terminal, **File > Open Folder**, and extensions retain **full real-user (milosvasic) host access by design** — a convenience view, not a security boundary. Real isolation would need a container / VM / chroot (out of scope per operator decision). Defense-in-depth (deploy stream): --disable-workspace-trust
 - /proxy/ blocked at the edge.
- **--auth none on loopback:8080 residual**. code-server runs unauthed on loopback; all auth is at the gate/edge. A local process reaching 127.0.0.1:8080 bypasses the gate — mitigated by scripts/harden-loopback.sh (a UID-scoped firewall rule the operator applies as root; the project never uses sudo).
- **Full in-browser editor interaction** (typing, file tree, command palette) is **OPERATOR-ATTENDED** until a browser-automation adapter lands (HCA-QA-UI-001); every HTTP/CLI-drivable journey is autonomously verified.
- **Post-0.0.3 additions now PASS (were in-flight/PENDING at Revision 2)**: the login-form clipboard copy/paste buttons (login_enhance.js, commit 8351e4c — LIVE 3/3 incl §11.4.170 CDP pixel proof), the VS Code Dark default theme (97a8c69 — config 3/3 + rendered-pixel proof), the popular-extensions matrix (97a8c69), the UI-driven marketplace journey (8351e4c — browse→search→Install-click, operator_attended on install-completion), and the rootless-Quadlet edge (now LIVE, active (running)).
- **CDP-harness honesty (§11.4.6)**: the §11.4.170 host-rendered pixel proofs for the login buttons and the dark theme are genuine but were captured in the runs cited in each row; the **latest** login_ui_auth L3 and theme_visual_auth runs honestly **SKIPPED** (Chromium Page.captureScreenshot timeout / workbench-never-rendered under the quiet matrix) — a skipped CDP run is never counted as the pixel-proof PASS, so the ledger cites the run where the driver reported ok:true / rc=0.
- **extensions_popular_auth is network-gated (§11.4.6)**: install/loadable/persist need a reachable Open VSX; the capability is proven live (run ...855046 installed 4/5 + the commit's direct 8-ext probe) and the network-independent config round-trip is green, but Open-VSX-unreachable runs install 0 — a bounded, honest **network-SKIP**, not a product FAIL and not a standing full-matrix-green claim.

- **Real, publicly-trusted Let's Encrypt** cannot be autonomously proven on a LAN box; the ACME code path is present (TLS_MODE=letsencrypt), the legacy local-CA suite now SKIPS as superseded, and the operator enables the public CA per docs/guides/TLS.md.
- **DOCX export.** Per §11.4.153(6) this doc class ships HTML + PDF + **DOCX**. This release regenerates all three for Status.md + Status_Summary.md via pandoc; the project has **no standing DOCX pipeline** in tests/run_all_types.sh yet — wiring a §11.4.60/§11.4.65 auto-export step for DOCX is a tracked follow-up.